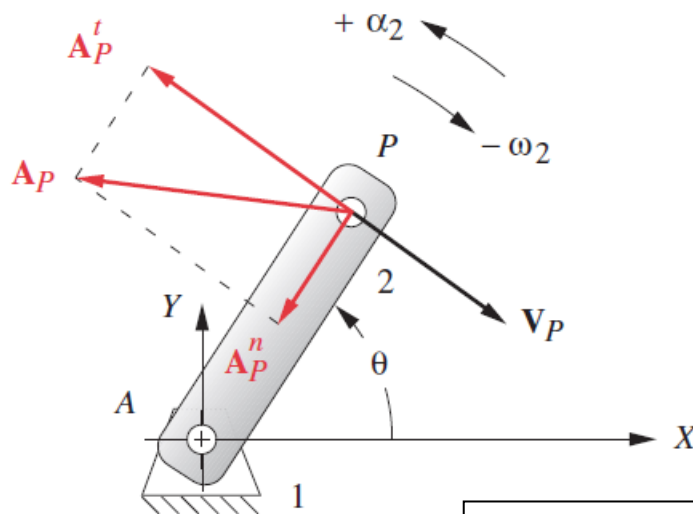


Linkage Analysis --Acceleration(Ch. 7)

- **Fundamentals**

- **Angular and linear accelerations** $\alpha = \frac{d\omega}{dt}$; $\mathbf{A} = \frac{d\mathbf{V}}{dt}$
- **Accelerations in a pure rotation**

Acceleration difference between those of Point P and Point A (rotational centre):



$$\mathbf{A}_{PA} = \frac{d\mathbf{V}_{PA}}{dt} = \frac{d(p\omega je^{j\theta})}{dt}$$

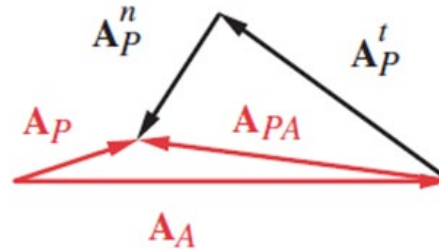
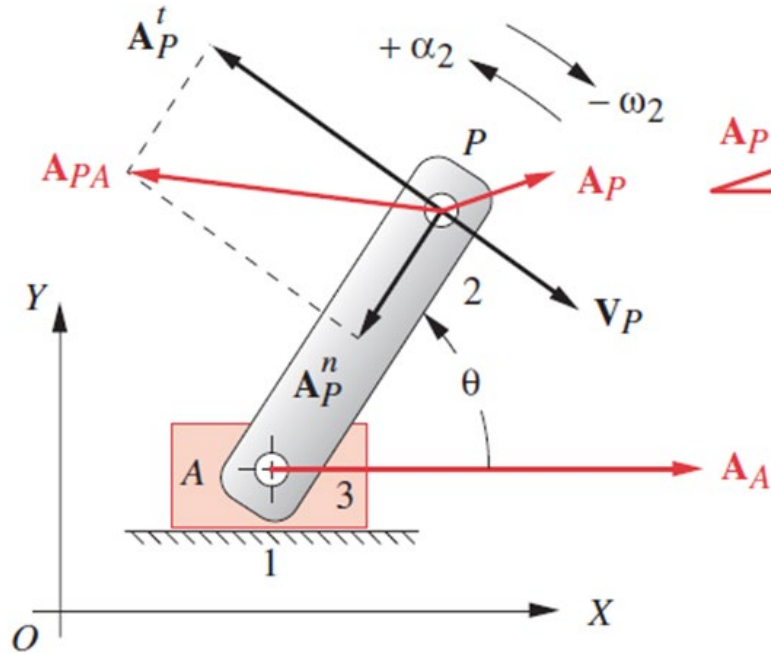
$$\mathbf{A}_{PA} = jp \left(e^{j\theta} \frac{d\omega}{dt} + \omega je^{j\theta} \frac{d\theta}{dt} \right)$$

$$\mathbf{A}_{PA} = p\alpha je^{j\theta} - p\omega^2 e^{j\theta}$$

$$\mathbf{A}_{PA} = \mathbf{A}_{PA}^t + \mathbf{A}_{PA}^n$$

Rotational acceleration = tangent part + centripetal part

– Accelerations in a complex motion (rotation cum translation; relative acceleration)

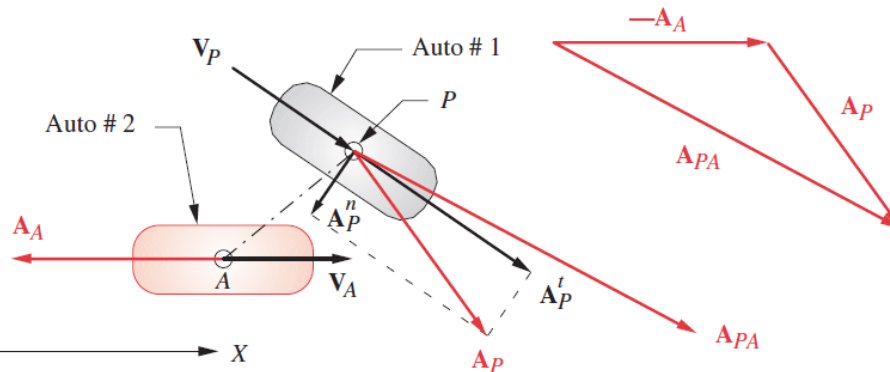


$$A_P = A_A + A_{PA}$$

$$(A_P^t + A_P^n) = (A_A^t + A_A^n) + (A_{PA}^t + A_{PA}^n)$$

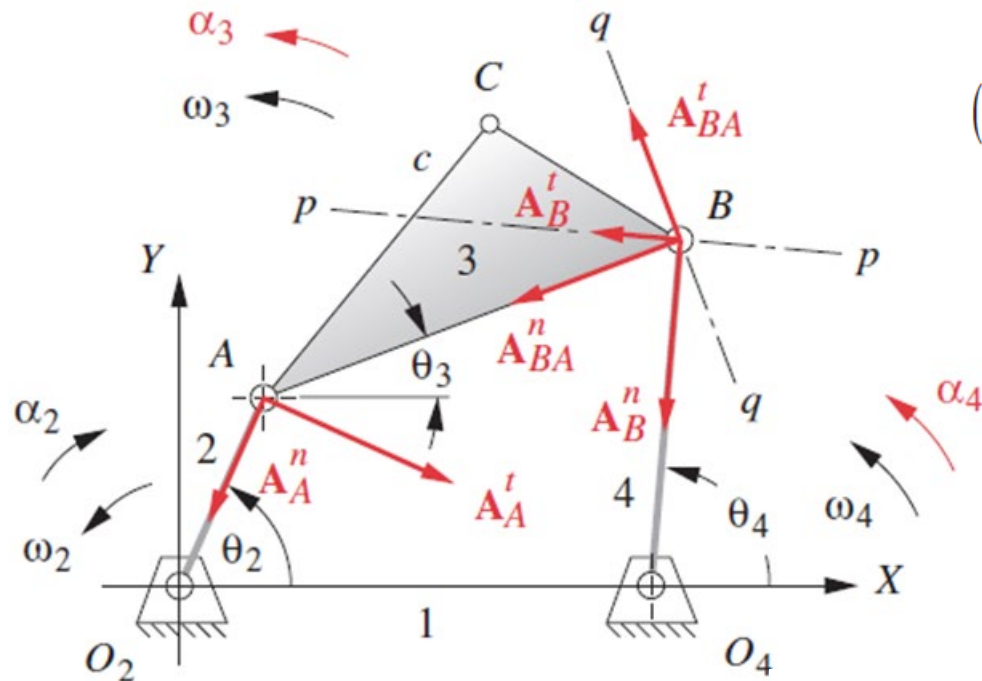
A_{PA} : Relative acceleration

Acceleration = translation component + rotation component



Example: Accelerations of two vehicles avoiding collision

- Acceleration analysis when the linkage is at a position -- graphic method

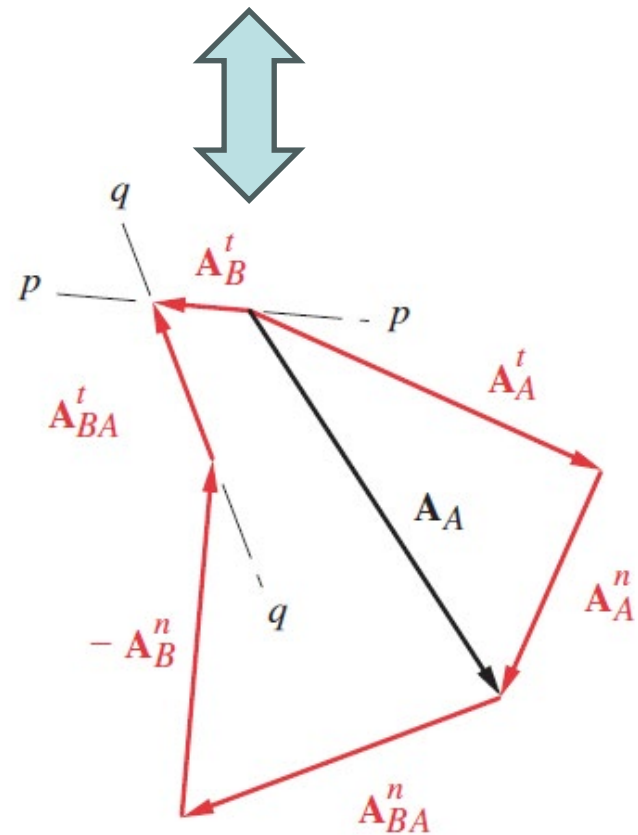


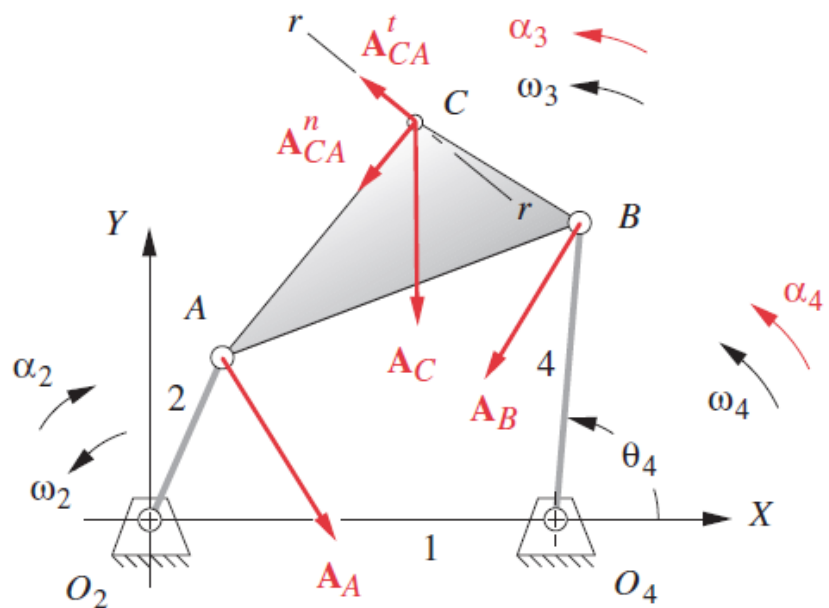
$$\mathbf{A}_B = \mathbf{A}_A + \mathbf{A}_{BA}$$

$$(\mathbf{A}_B^t + \mathbf{A}_B^n) = (\mathbf{A}_A^t + \mathbf{A}_A^n) + (\mathbf{A}_{BA}^t + \mathbf{A}_{BA}^n)$$

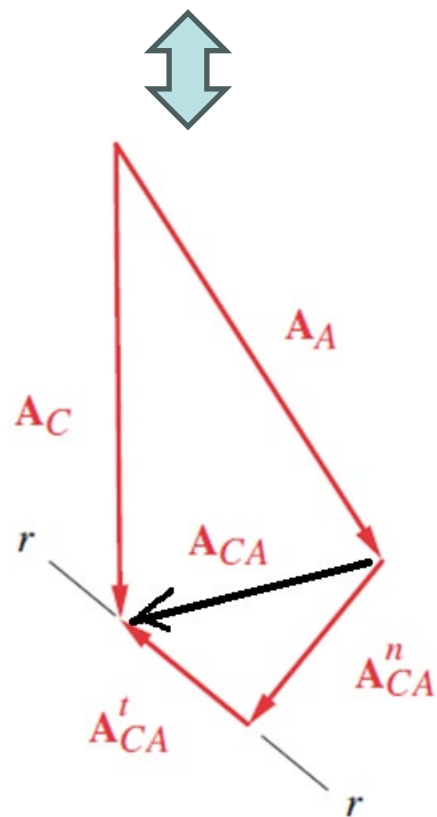
$$\alpha_4 = \frac{A_B^t}{BO_4}$$

$$\alpha_3 = \frac{A_{BA}^t}{BA}$$





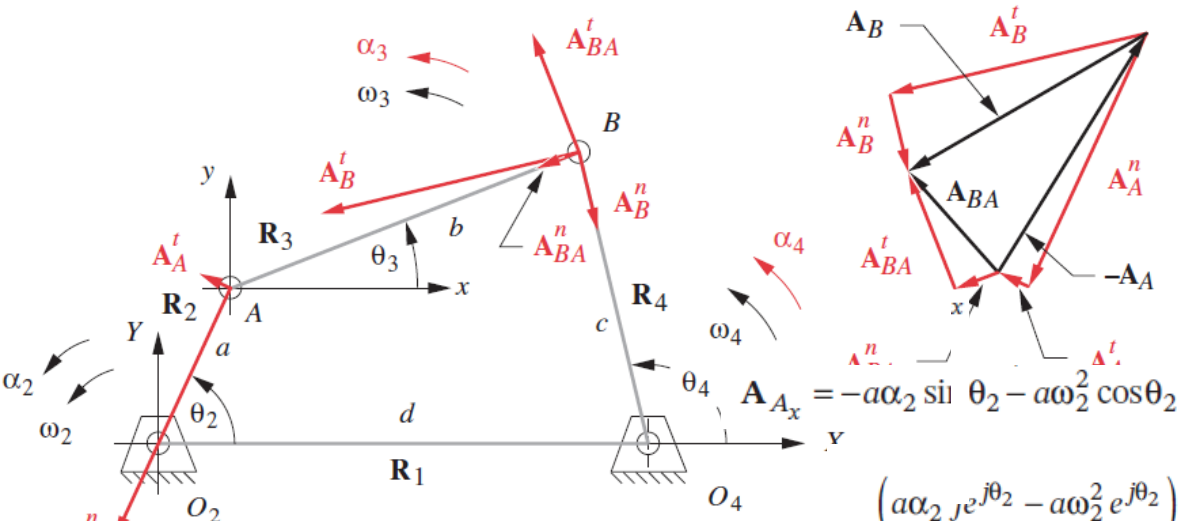
$$\mathbf{A}_C = \mathbf{A}_A + \mathbf{A}_{CA}$$



$$A_{CA}^t = c\alpha_3$$

$$A_{CA}^n = c\omega_3^2$$

- Acceleration analysis when the linkage is at any position – Four-bar linkage



With velocity analysis:

$$ja\omega_2 e^{j\theta_2} + jb\omega_3 e^{j\theta_3} - jc\omega_4 e^{j\theta_4} = 0$$

Differentiate the above with time

$$\mathbf{A}_A + \mathbf{A}_{BA} - \mathbf{A}_B = 0$$

$$(a\alpha_2 j e^{j\theta_2} - a\omega_2^2 e^{j\theta_2}) + (b\alpha_3 j e^{j\theta_3} - b\omega_3^2 e^{j\theta_3}) - (c\alpha_4 j e^{j\theta_4} - c\omega_4^2 e^{j\theta_4}) = 0$$

$$a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2 + b\alpha_3 \cos\theta_3 - b\omega_3^2 \sin\theta_3 - c\alpha_4 \cos\theta_4 + c\omega_4^2 \sin\theta_4 = 0$$

$$-a\alpha_2 \sin\theta_2 - a\omega_2^2 \cos\theta_2 - b\alpha_3 \sin\theta_3 - b\omega_3^2 \cos\theta_3 + c\alpha_4 \sin\theta_4 + c\omega_4^2 \cos\theta_4 = 0$$

$$\alpha_3 = \frac{CD - AF}{AE - BD}$$

$$\alpha_4 = \frac{CE - BF}{AE - BD}$$

$$A = c \sin\theta_4$$

$$B = b \sin\theta_3$$

$$C = a\alpha_2 \sin\theta_2 + a\omega_2^2 \cos\theta_2 + b\omega_3^2 \cos\theta_3 - c\omega_4^2 \cos\theta_4$$

$$D = c \cos\theta_4$$

$$E = b \cos\theta_3$$

$$F = a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2 - b\omega_3^2 \sin\theta_3 + c\omega_4^2 \sin\theta_4$$

$$\mathbf{A}_A = a\alpha_2 (-\sin\theta_2 + j \cos\theta_2) - a\omega_2^2 (\cos\theta_2 + j \sin\theta_2)$$

$$\mathbf{A}_{Ax} = -a\alpha_2 \sin\theta_2 - a\omega_2^2 \cos\theta_2$$

$$\mathbf{A}_{Ay} = a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2$$

$$\mathbf{A}_{BA} = b\alpha_3 (-\sin\theta_3 + j \cos\theta_3) - b\omega_3^2 (\cos\theta_3 + j \sin\theta_3)$$

$$\mathbf{A}_{BAx} = -b\alpha_3 \sin\theta_3 - b\omega_3^2 \cos\theta_3$$

$$\mathbf{A}_{BAy} = b\alpha_3 \cos\theta_3 - b\omega_3^2 \sin\theta_3$$

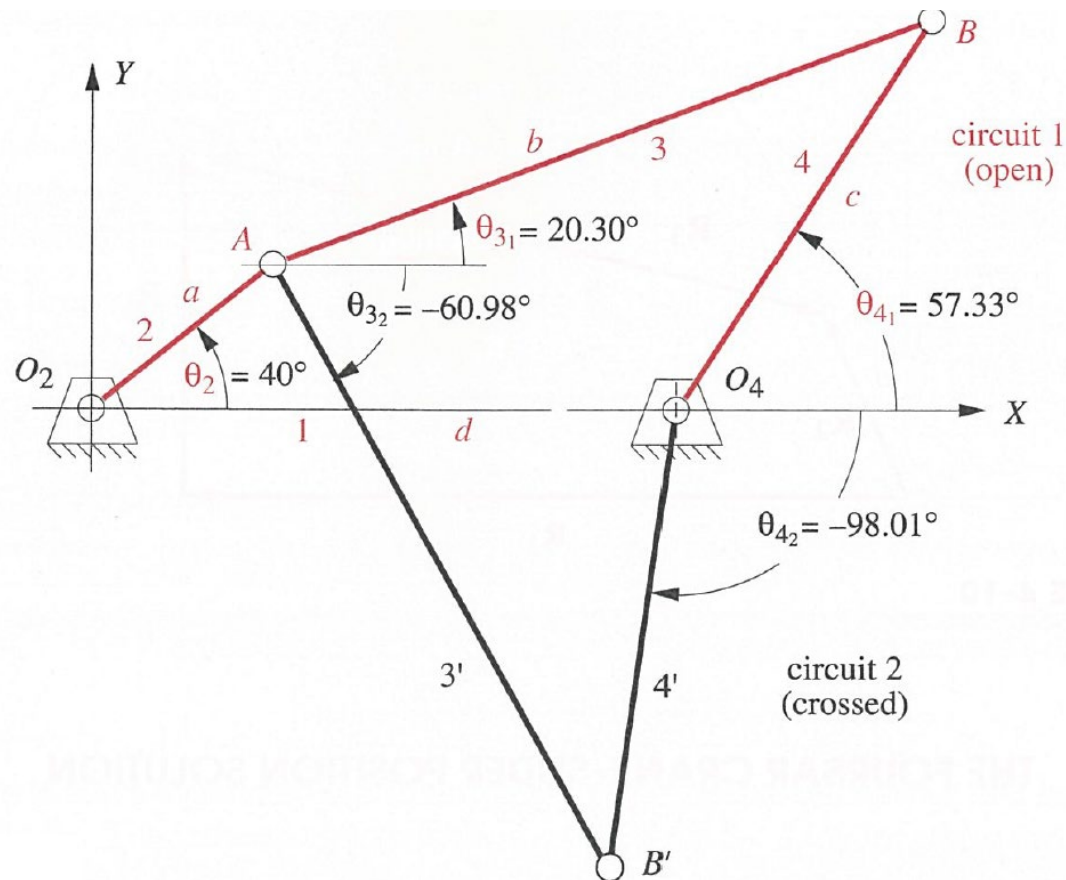
$$\mathbf{A}_B = c\alpha_4 (-\sin\theta_4 + j \cos\theta_4) - c\omega_4^2 (\cos\theta_4 + j \sin\theta_4)$$

$$\mathbf{A}_{Bx} = -c\alpha_4 \sin\theta_4 - c\omega_4^2 \cos\theta_4$$

$$\mathbf{A}_{By} = c\alpha_4 \cos\theta_4 - c\omega_4^2 \sin\theta_4$$

Example 1

Given a fourbar linkage with the link lengths $L_1 = d = 100$ mm, $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, $L_4 = c = 80$ mm. For $\theta_2 = 40^\circ$, $\omega_2 = 25$ rad/sec, and $\alpha_2 = 15$ rad/sec² find the values of α_3 and α_4 , A_A , A_{BA} , and A_B for the open circuit of the linkage. Use the angles and angular velocities found for the same linkage and position in Example 6-7. $\omega_3 = -4.121$ and $\omega_4 = 6.998$ rad/sec.



Solution

$$A = c \sin \theta_4 = 80 \sin 57.325^\circ = 67.340$$

$$B = b \sin \theta_3 = 120 \sin 20.298^\circ = 41.628$$

$$C = a\alpha_2 \sin \theta_2 + a\omega_2^2 \cos \theta_2 + b\omega_3^2 \cos \theta_3 - c\omega_4^2 \cos \theta_4 = 19332.98$$

$$D = c \cos \theta_4 = 80 \cos 57.325^\circ = 43.190$$

$$E = b \cos \theta_3 = 120 \cos 20.298^\circ = 112.548 \quad \text{---}$$

$$F = a\alpha_2 \cos \theta_2 - a\omega_2^2 \sin \theta_2 - b\omega_3^2 \sin \theta_3 + c\omega_4^2 \sin \theta_4 = -13\,019.25$$

$$\alpha_3 = \frac{CD - AF}{AE - BD} = 296.089 \text{ rad/sec}^2$$

$$\alpha_4 = \frac{CE - BF}{AE - BD} = 470.134 \text{ rad/sec}^2$$

$$\mathbf{A}_{A_x} = -a\alpha_2 \sin \theta_2 - a\omega_2^2 \cos \theta_2 = -19.537 \text{ m/sec}^2$$

$$\mathbf{A}_{A_y} = a\alpha_2 \cos \theta_2 - a\omega_2^2 \sin \theta_2 = -15.617 \text{ m/sec}^2$$

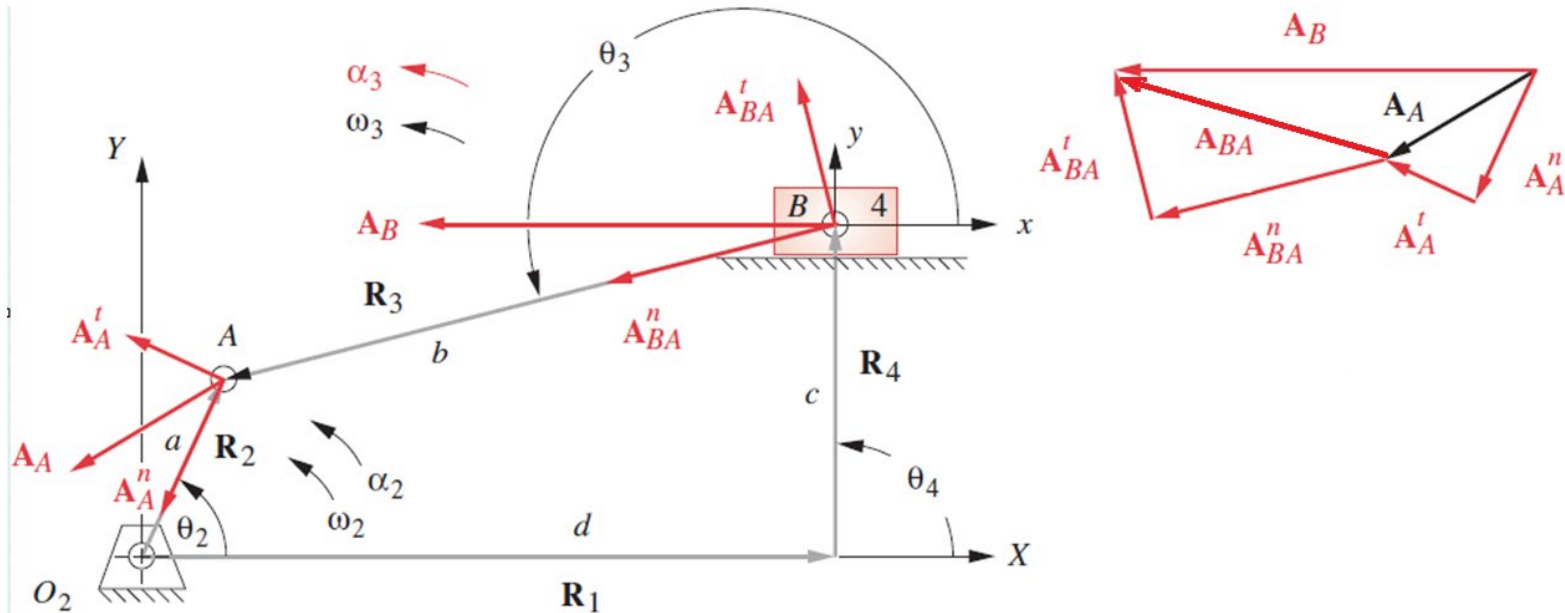
$$\mathbf{A}_{BA_x} = -b\alpha_3 \sin \theta_3 - b\omega_3^2 \cos \theta_3 = -14\,237 \text{ m/sec}^2$$

$$\mathbf{A}_{BA_y} = b\alpha_3 \cos \theta_3 - b\omega_3^2 \sin \theta_3 = 32.617 \text{ m/sec}^2$$

$$\mathbf{A}_{B_x} = -c\alpha_4 \sin \theta_4 - c\omega_4^2 \cos \theta_4 = -33.774 \text{ m/sec}^2$$

$$\mathbf{A}_{B_y} = c\alpha_4 \cos \theta_4 - c\omega_4^2 \sin \theta_4 = 17.007 \text{ m/sec}^2$$

– Crank Slider



With velocity analysis, $ja\omega_2 e^{j\theta_2} - jb\omega_3 e^{j\theta_3} - \dot{d} = 0$

Differentiate the above with time, $(a\alpha_2 je^{j\theta_2} - a\omega_2^2 e^{j\theta_2}) - (b\alpha_3 je^{j\theta_3} - b\omega_3^2 e^{j\theta_3}) - \ddot{d} = 0$

$$\mathbf{A}_A - \mathbf{A}_{AB} - \mathbf{A}_B = 0$$

$$-a\alpha_2 \sin \theta_2 - a\omega_2^2 \cos \theta_2 + b\alpha_3 \sin \theta_3 + b\omega_3^2 \cos \theta_3 - \ddot{d} = 0$$

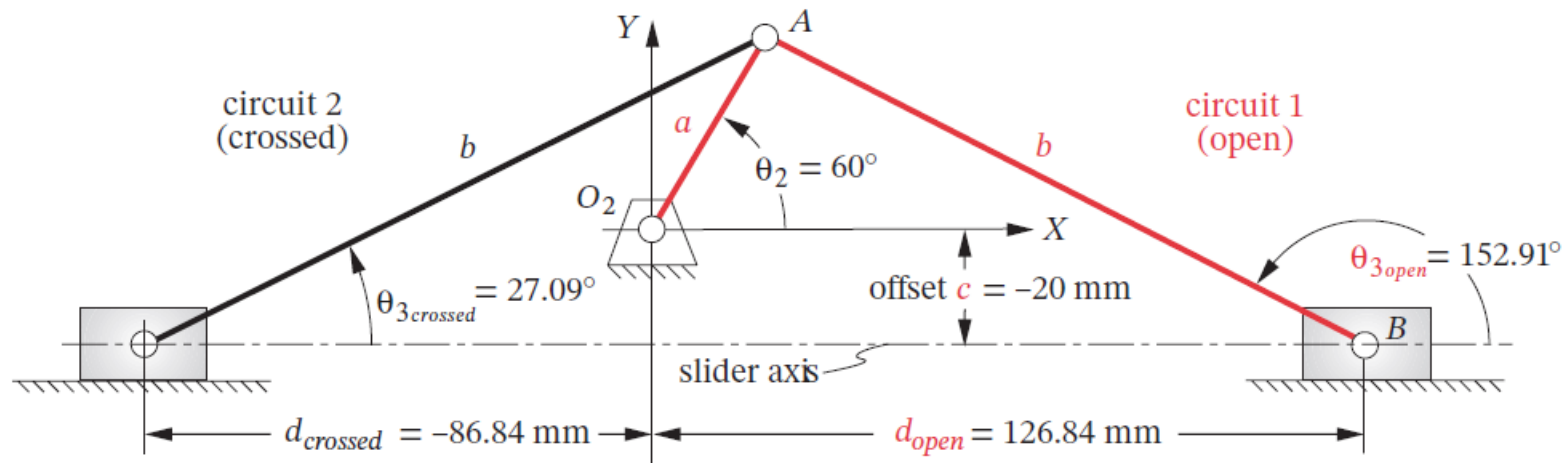
$$a\alpha_2 \cos \theta_2 - a\omega_2^2 \sin \theta_2 - b\alpha_3 \cos \theta_3 + b\omega_3^2 \sin \theta_3 = 0$$

$$\alpha_3 = \frac{a\alpha_2 \cos \theta_2 - a\omega_2^2 \sin \theta_2 + b\omega_3^2 \sin \theta_3}{b \cos \theta_3}$$

$$\ddot{d} = -a\alpha_2 \sin \theta_2 - a\omega_2^2 \cos \theta_2 + b\alpha_3 \sin \theta_3 + b\omega_3^2 \cos \theta_3$$

Example 2

Given a fourbar crank-slider linkage with the link lengths $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, *offset* = $c = -20$ mm. For $\theta_2 = 60^\circ$, $\omega_2 = -30$ rad/sec, and $\alpha_2 = 20$ rad/sec², find α_3 and linear acceleration of the slider for the open circuit. Use the angles, positions, and angular velocities found for the same linkage in Examples 4-2 and 6-8. $\omega_3 = 5.616$ rad/sec $\dot{d} = 1346$ mm/sec



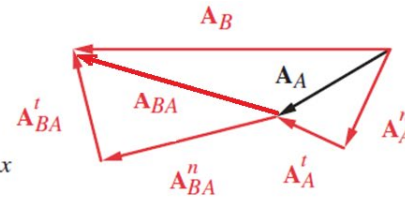
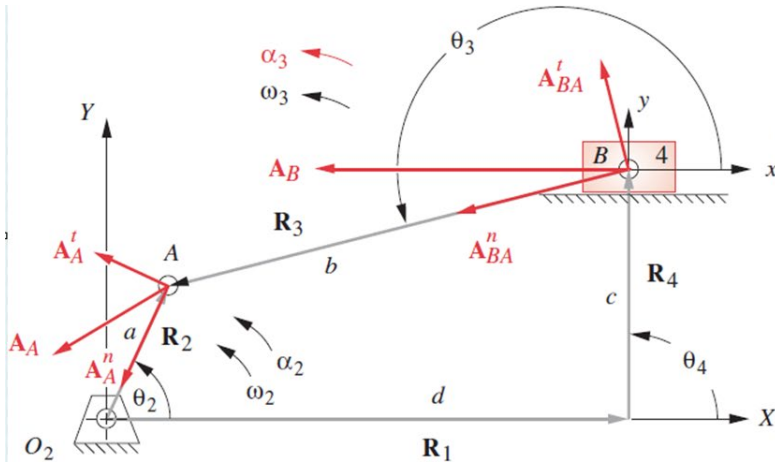
Solution

$$\alpha_3 = \frac{a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2 + b\omega_3^2 \sin\theta_3}{b \cos\theta_3} = 271.94 \text{ rad/sec}^2$$

$$\ddot{d} = -a\alpha_2 \sin\theta_2 - a\omega_2^2 \cos\theta_2 + b\alpha_3 \sin\theta_3 + b\omega_3^2 \cos\theta_3 = -7.203 \text{ m/sec}^2$$

– Slider Crank

Same acceleration equations, but different input/output !



$$-a\alpha_2 \sin\theta_2 - a\omega_2^2 \cos\theta_2 + b\alpha_3 \sin\theta_3 + b\omega_3^2 \cos\theta_3 - \ddot{d} = 0$$

$$a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2 - b\alpha_3 \cos\theta_3 + b\omega_3^2 \sin\theta_3 = 0$$



$$\alpha_3 = \frac{a\alpha_2 \cos\theta_2 - a\omega_2^2 \sin\theta_2 + b\omega_3^2 \sin\theta_3}{b \cos\theta_3}$$

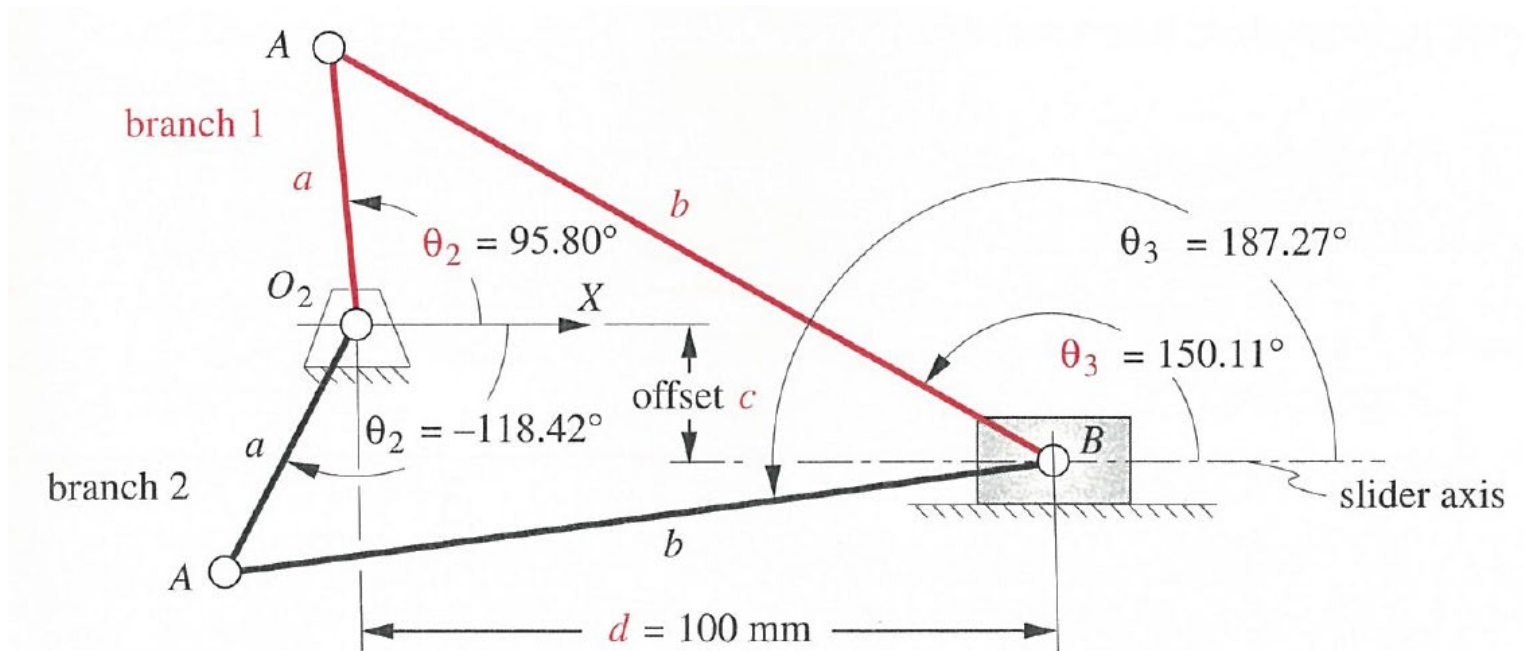
$$\alpha_2 = \frac{a\omega_2^2 (\cos\theta_2 \cos\theta_3 + \sin\theta_2 \sin\theta_3) - b\omega_3^2 + \ddot{d} \cos\theta_3}{a(\cos\theta_2 \sin\theta_3 - \sin\theta_2 \cos\theta_3)}$$

Example 3

Given a fourbar slider-crank linkage with the link lengths $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, $offset = c = -20$ mm. For $d = 100$ mm and $\dot{d} = 900$ mm/sec², find α_2 and α_3 for both branches of one circuit of the linkage. Use the angles and angular velocities found for the same linkage in Example 4-3 and Example 6-9, respectively.

$$\omega_{2_1} = -32.023 \text{ and } \omega_{3_1} = -1.244 \text{ rad/sec}$$

$$\omega_{2_2} = 36.64 \text{ and } \omega_{3_2} = 5.86 \text{ rad/sec}$$



Solution

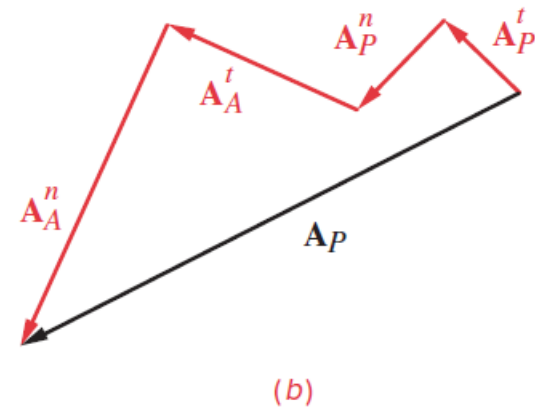
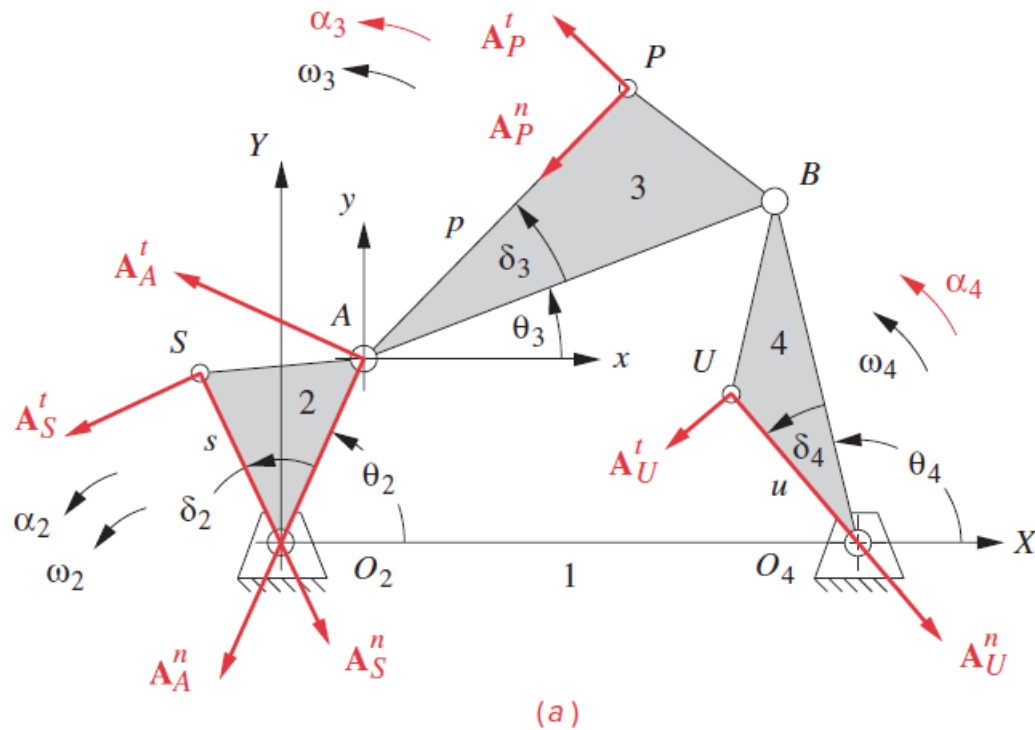
$$\alpha_{2_1} = \frac{a\omega_{2_1}^2 (\cos\theta_{2_1} \cos\theta_{3_1} + \sin\theta_{2_1} \sin\theta_{3_1}) - b\omega_{3_1}^2 + \ddot{d} \cos\theta_{3_1}}{a(\cos\theta_{2_1} \sin\theta_{3_1} - \sin\theta_{2_1} \cos\theta_{3_1})} = 706.753 \text{ rad/sec}^2$$

$$\alpha_{3_1} = \frac{a\alpha_{2_1} \cos\theta_{2_1} - a\omega_{2_1}^2 \sin\theta_{2_1} + b\omega_{3_1}^2 \sin\theta_{3_1}}{b \cos\theta_{3_1}} = 418.804 \text{ rad/sec}^2$$

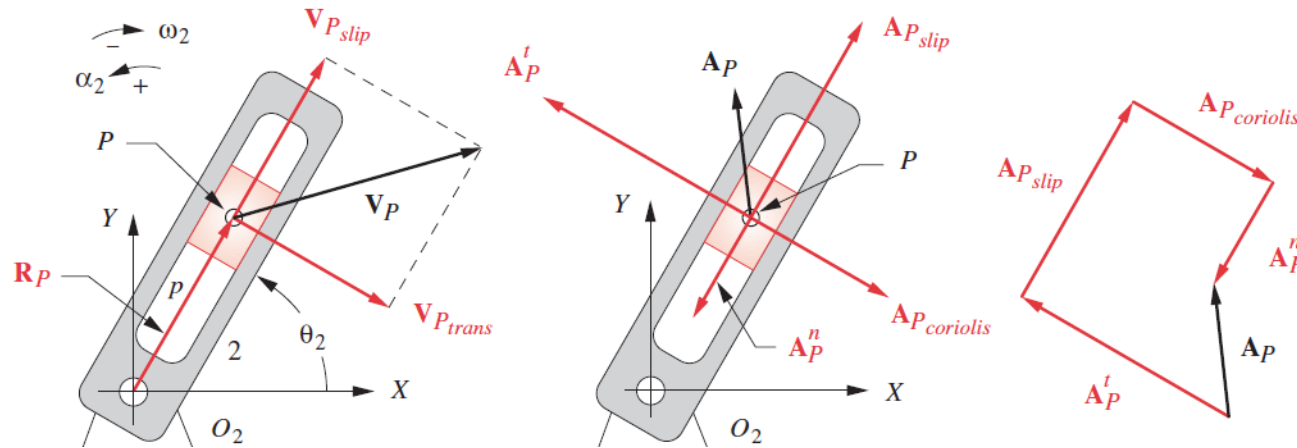
$$\alpha_{2_2} = \frac{a\omega_{2_2}^2 (\cos\theta_{2_2} \cos\theta_{3_2} + \sin\theta_{2_2} \sin\theta_{3_2}) - b\omega_{3_2}^2 + \ddot{d} \cos\theta_{3_2}}{a(\cos\theta_{2_2} \sin\theta_{3_2} - \sin\theta_{2_2} \cos\theta_{3_2})} = -809.801 \text{ rad/sec}^2$$

$$\alpha_{3_2} = \frac{a\alpha_{2_2} \cos\theta_{2_2} - a\omega_{2_2}^2 \sin\theta_{2_2} + b\omega_{3_2}^2 \sin\theta_{3_2}}{b \cos\theta_{3_2}} = -521.852 \text{ rad/sec}^2$$

- Acceleration of any point in a linkage



- Coriolis acceleration (of a point moving relative to a rotational link)



$$\mathbf{R}_P = p e^{j\theta_2} \Rightarrow \mathbf{V}_P = p\omega_2 j e^{j\theta_2} + \dot{p} e^{j\theta_2} \Rightarrow \mathbf{A}_P = p\alpha_2 j e^{j\theta_2} - p\omega_2^2 e^{j\theta_2} + 2\dot{p}\omega_2 j e^{j\theta_2} + \ddot{p} e^{j\theta_2}$$

$$\mathbf{V}_P = \mathbf{V}_{P_{trans}} + \mathbf{V}_{P_{slip}}$$

$$\mathbf{A}_P = \mathbf{A}_{P_{tangential}} + \mathbf{A}_{P_{normal}} + \mathbf{A}_{P_{coriolis}} + \mathbf{A}_{P_{slip}}$$